

Mechanical Inspections - LEEA COPSULE

Extract

Pertinent information for Stage Electrics: integrated lifting systems, lifting appliances, supporting structures and lifting accessories

Source reviewed: Lifting Equipment Engineers Association, Code of Practice for the Safe Use of Lifting Equipment, Edition 9.1, January 2024. References below are to section numbers in that document.

Executive summary

- The document does not appear to use the phrase "Integrated Lifting Systems" directly. This extract treats it as installed lifting systems made up of appliances, supporting structures, runways/tracks, trolleys, beam clamps, hoists/winches, controls and associated lifting accessories.
- Inspection should be applied to the whole lifting arrangement, not just the hoist or single asset.
- LEEA defines three inspection levels: pre-use inspection, interim inspection and thorough examination. Thorough examination is the formal competent-person inspection that typically results in a report.
- LEEA recommends maximum intervals of 12 months for goods lifting appliances and 6 months for lifting accessories and people-carrying equipment, unless a written scheme of examination justifies otherwise.
- Routine overload testing is generally not recommended unless required by legislation, manufacturer instructions or exceptional circumstances such as significant modification or repair.

1. Core inspection framework

Inspection level	Meaning	Practical implication for Stage Electrics
Pre-use inspection	Basic check before use, normally by the operator, for obvious damage, missing/incorrect marking, abnormal noise, deformation or functional change.	Operators/customers should be trained to complete and report defects before use.
Interim inspection	Risk-based inspection between thorough examinations, focused on critical components that may become dangerous before the next thorough examination.	Important for high-use venues, theatres, touring, production and hire environments. Can be combined with planned maintenance.
Thorough examination	Careful and critical visual examination by a Competent Person, supplemented where appropriate by measurement, functional testing, proof-load testing or NDT.	Formal inspection likely to be reported by Stage Electrics. It also checks whether the inspection/maintenance regime is adequate.

Reference: COPSULE Section 1.2.3 and Section 1.7.

2. Competent Person and reports

- A Competent Person must have the necessary practical and theoretical knowledge, experience, training, skill and ability for the specific inspection duty. LEEA notes that competency can be a corporate responsibility.

Source: LEEA Code of Practice for the Safe Use of Lifting Equipment, Edition 9.1, January 2024

- A Report of Thorough Examination should state the results, defects found, or that the item is fit for continued use. Where tests are performed, the report should include test details.
- Reports of thorough examination must be retained as part of the lifting equipment records.

Recommended report contents
Identification mark / asset number
WLL / SWL / rated capacity
Manufacturer, model and serial number where applicable
Location or installation reference
Date of examination and next due date / scheme reference
Defects, actions required and risk category where used
Fit for continued use statement
Details of tests carried out, if any
Cross-reference to historical records, repairs, modifications and previous examinations

Reference: COPSULE Sections 1.2.1, 1.2.15 and 1.7.6.

3. Integrated lifting systems / installed lifting systems

For installed or integrated systems, the inspection scope may include:

- bridge and gantry cranes
- hand chain hoists and chain lever hoists
- wire-rope gripping machines
- power-operated hoists
- lifting winches
- travelling girder trolleys
- beam clamps
- slewing jib cranes
- runways
- mobile supporting structures
- jacks
- supporting steelwork, lifting points and anchorage points
- power feed and control systems
- below-hook lifting accessories
- Supporting structures and lifting points require more than proof-load testing. Dedicated supporting structures such as runways should have a theoretical check by a structural engineer or suitably qualified person, and should be constructed, tested, certified and periodically thoroughly examined.
- Other lifting points should also have theoretical checks, written confirmation, properly designed/tested/certified connections and periodic thorough examination.
- Dedicated supporting structures and approved lifting points should be marked with SWL. LEEA recommends adding MANUAL or POWER as appropriate, because power-operated hoists impose higher dynamic loading allowances.

Reference: COPSULE Sections 2-13 and Appendix 1.4.

4. Installation, commissioning, modification and repair

- Installed systems should be installed by appropriately qualified personnel and checked against design criteria and dimensions.

- Post-installation checks may include controls, limit switches, emergency stops, electrical safety, full-load operation, hoist/travel speeds and overload testing of the crane/supporting structure where appropriate.
- After installation, significant modification or repair, the employer must obtain a report of thorough examination confirming the equipment has been properly installed and is safe to use.
- Repairs to load-bearing parts using certified spare parts should be followed by thorough examination and record updates. Repairs with uncertified spare parts require verification to the original product specification followed by thorough examination.

Reference: COPSULE Sections 1.7.4, 1.7.5 and relevant equipment sections, including bridge and gantry crane guidance.

5. Inspection intervals

Equipment / circumstance	Typical LEEA maximum interval or trigger
Goods lifting appliances: cranes, hoists, winches, trolleys, runways, jibs and mobile supporting structures	12 months, unless a written scheme of examination specifies otherwise
Lifting accessories / below-hook equipment	6 months, unless a written scheme of examination specifies otherwise
People-carrying lifting equipment	6 months
After installation, substantial repair, modification, collision or other exceptional event	Before return to service / after the event as required by the Competent Person

Reference: COPSULE Section 1.2.3.3 and Section 1.7.

6. Power-operated hoists, winches and controls

- Power-operated hoists are self-sustaining and may be used in temporary or permanent applications. They may be suspended, built into trolleys, fixed suspension mounted, foot mounted, or used on runways, jibs and mobile supporting structures.
- Winches are particularly relevant to theatre and fixed-position lifting. LEEA notes they may be built into structures or arranged with diverter pulleys, can be remote from the load, and are used in theatres as well as construction, vehicle-mounted lifting arrangements, factories and stores.
- Remote controls should be maintained, especially seals around buttons/joysticks and casing condition, because contamination or water ingress can cause false commands. Controls should have a designated storage place and the stop/main contactor should be used before leaving controls unattended.

Reference: COPSULE Appendix 1.6 and Section 7 guidance on winches.

7. Mobile supporting structures

- A mobile supporting structure is not, by itself, a lifting appliance. It becomes an effective lifting structure only when fitted with a hoist, hoist/trolley or similar appliance.
- Standard mobile supporting structures are generally not intended to be moved under load. Written manufacturer confirmation is required if movement under load is intended.
- The associated hoist, trolley, beam clamp, winch or pulley arrangement must also be assessed under its own relevant section.

Reference: COPSULE Section 12 and Appendix 1.6.

8. Lifting accessories / below-hook equipment

LEEA defines a lifting accessory as equipment used to attach the load to the lifting appliance. This includes rigging equipment, lifting gear and lifting tackle. Relevant categories include:

- chain slings
 - wire rope slings
 - flat woven webbing slings
 - man-made fibre roundslings
 - fibre rope slings
 - shackles
 - eyebolts
 - lifting beams, spreaders and frames
 - plate handling clamps
 - rigging screws and turnbuckles
 - barrel lifters
 - lifting forks
 - magnetic lifters
 - vacuum lifters
- Most lifting accessories follow the same inspection pattern: pre-use inspection, risk-based interim inspection where required, maintenance to manufacturer instructions and site conditions, and thorough examination by a Competent Person at a maximum interval of 6 months unless a written scheme states otherwise.
 - Items incorporated into the load still need to be considered in the lift plan for adequate strength, integrity, inspection and freedom from defect.

Reference: COPSULE Sections 14-27 and Section 1.2.4.

9. Key inspection defects to watch for

- missing, illegible or incorrect SWL/WLL/rated-capacity markings
- missing or illegible identification
- distortion, bending, elongation or deformation
- cracks, nicks, gouges, corrosion or fatigue signs
- wear of pins, shackles, eyes, hooks, links, holes and load-bearing parts
- incorrect or damaged pins or threads
- damaged, contaminated or worn pads/jaws on clamps, vacuum lifters and specialist accessories
- damaged controls, warning lights, indicators, limit devices or emergency stops
- loose or incorrectly fitted bolts and fixings
- worn, contaminated or slipping trolley wheels/runway faces
- damaged load chain, wire rope, textile sling fabric or fittings
- chemical, heat, UV, abrasion, cutting or soiling damage to textile/fibre equipment
- missing or ineffective end stops
- unsupported or unverified structural connections
- any change in function, abnormal noise or performance

For lifting beams, spreaders and frames, specific checks include lifting eyes, bolts, load attachment points, shackle holes, bolted connections, beam distortion, hollow-section dents/local buckling, corrosion, cracks or distortion in welds, and confirmation that markings match documentation.

Reference: COPSULE Sections 1.6, 1.7 and relevant equipment sections, particularly Section 21 for lifting beams/spreaders/frames.

10. Testing and overload testing

- Testing is not limited to proof-load testing. LEEA includes proof-load tests, operational tests at lower loads and non-destructive testing within the term testing.
- Verification may include proof-load tests, sample break tests, NDT, calculation, measurement and thorough examination.
- LEEA generally does not recommend routine overload testing unless required by legislation, manufacturer instructions or exceptional circumstances such as significant modification or repair.
- Reasons include repeated overloads causing deterioration, many defects not being revealed by overload testing, and defects such as fatigue cracking, deformation, tears, cuts or fraying potentially being worsened without being identified by the test.

Reference: COPSULE Sections 1.2.14, 1.2.19 and equipment-specific testing guidance.

11. Marking, storage and handling

- Equipment that has been verified and passed thorough examination should be marked with WLL/SWL or rated capacity, an unambiguous identification linked to documentation, and any other required marks.
- If required marking is obliterated or illegible, the equipment should be withdrawn from service and referred to a Competent Person for re-marking, re-verification and/or re-marking as needed.
- Storage should reduce damage and deterioration. In general, equipment should be stored dry, free from injurious pollution, away from extreme temperatures, and protected where it has exposed threads or machined bearing surfaces such as eyebolts and shackles.
- Wet or contaminated equipment should be cleaned and dried before storage.

Reference: COPSULE Sections 1.6.1 and 1.6.2.

12. Operational safety points to pull into guidance

- determine load weight and centre of gravity before lifting
- use load sensing where weight or individual component loading is uncertain
- use equipment only in accordance with manufacturer instructions
- do not overload by load weight, sling angle or slinging method
- ensure load balance, stability and security
- complete pre-use inspection before lifting
- account for obstacles such as power lines, pipework and other lifting operations
- prepare the landing area and confirm it can take the load
- secure loose parts of the load
- use clear communication and agreed signals
- keep people clear of suspended loads; if unavoidable, provide secondary safeguards
- make a trial lift first to check balance, stability and security
- do not trap slings under loads and never drag slings out with the lifting appliance
- return equipment to proper storage after use

- For multi-point lifting, take steps to ensure that no item of equipment or lifting point exceeds its SWL. Load sensors, careful calculation, linked control and load limiting/monitoring may be required depending on the operation.

Reference: COPSULE Section 1.8 and Appendices 1.2, 1.3 and 1.5.

13. Training relevance

- Operatives must be trained in the specific equipment they use and should use only equipment for which they have received training or instruction.
- Training should cover markings and limitations, daily checks, pre-use inspection, defect reporting, correct attachment of loads/slings/accessories, trial lifts, lifting/lowering procedures, signals, withdrawal from service, storage and seeking advice when in doubt.
- Inspection findings may indicate a training issue, not only an equipment issue. Defects found at thorough examination should lead to root-cause assessment and measures to prevent recurrence, such as increased inspection or operator training.

Reference: COPSULE Section 1.9 and equipment-specific inspection/maintenance guidance.

Practical takeaway for Stage Electrics

Treat every installation as a complete lifting arrangement, not a single asset. For integrated lifting systems, the inspection scope should consider the lifting appliance, supporting structure, runway/track, trolley, beam clamps or fixings, power/control system, limits, emergency stops, markings, documentation, maintenance history, lifting accessories and the user's inspection regime.

- Default intervals: 12 months for goods lifting appliances and supporting systems; 6 months for lifting accessories and people-carrying equipment; sooner after installation, repair, modification, collision or exceptional events.
- Default action on serious defects: withdraw from service, refer to a Competent Person, record findings, investigate root cause, remedy and prevent recurrence before return to service.